
When the Cross-Silo Federation Goes Offline: Continual Learning for Site Onboarding with Limited Unlabeled Data

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Abstract

1 An organization often holds too little labeled data to train a model that general-
2 izes, and the records that would supply the rest sit with organizations that cannot
3 release them. Cross-silo federated learning offers a way through, since partici-
4 pants exchange model parameters rather than records, but it ordinarily settles two
5 aspects of the arrangement in advance, the participating sites and the classes the
6 model can predict, and deployment can breach both. A new site joins after train-
7 ing, once the established sites have finished their engagement and gone offline,
8 and its records arrive unlabeled, mixing conditions the model already recognizes
9 with conditions no participant has observed. We present an autonomous three-
10 stage procedure that expands the model entirely at the joining site: reconstruction
11 experts screen for novelty, clustering separates the flagged records into candidate
12 conditions, and class means describe the old classes, all inside one shared repre-
13 sentation. Those classes were learned from records that never leave their owners,
14 so the usual defenses against forgetting are unavailable, and the procedure sup-
15 plies the evidence they would have carried from either of two dissimilar sources,
16 prototypes held by the federation or records held by the joining site. On a real
17 industrial condition-monitoring dataset, run end to end with no label consulted,
18 either source holds old-class accuracy at 0.868 or above with forgetting at most
19 0.063, and the two differ by 0.021, less than a single test record is worth, so a
20 configuration can be chosen by the disclosure it permits rather than the accuracy
21 it delivers. Both keep old- and new-class accuracy in balance where every alter-
22 native we measure gives up one task for the other, and both retain more of the
23 old classes than distillation- and regularization-based baselines. The balance still
24 holds with only 6 labeled records per arriving condition and 3 retained per old
25 class.

1 Introduction

26 A single organization often holds too little labeled data to train a model that generalizes well, while
27 the records that would supply the rest sit with other organizations that cannot release them, since
28 regulation, contracts and commercial sensitivity keep patient records, transaction histories and pro-
29 cess measurements inside the firm that collected them. Federated learning offers a way through.
30 The participants exchange model parameters instead of records, so each contributes what its data
31 taught without disclosing the data itself [11, 18]. Most deployments then settle the arrangement at
32 the outset in two respects. The set of participating sites is fixed, and so is the set of classes the model
33 can predict.
34

Federated continual learning relaxes these commitments [6, 26], though usually under assumptions that a site joining a model already in production is unlikely to meet. Deployment can breach both at once. A new plant, hospital or business unit joins after the model is in service and brings operating conditions the federation never observed. By then the sites that trained the model may have completed their engagement and gone offline, leaving the joining site what the base session produced rather than a federation it can convene. Class-incremental learning allows the output space to grow, but training on the new conditions also degrades accuracy on the classes learned earlier, an effect known as catastrophic forgetting [17].

Forgetting is harder to counter inside a federation. Centralized continual learning counters forgetting in three broad ways, and this setting removes or weakens each. Replay stores a few examples of the old classes and mixes them into the new training data [20]. It is usually the most reliable and the first to fail here, because the old classes were learned from records that belong to other organizations and never left them. Regularization anchors the parameters to the previous solution using a curvature estimate such as a Fisher information [12]; it survives, but only if that estimate is computed where the old data are and shipped before the sites disconnect. Distillation keeps the new model’s outputs close to a frozen copy of the old one [14], and a joining site can apply it only to the records it holds, so the old classes are anchored through data drawn from a different distribution.

What a late-joining site can and cannot use. Three constraints act together at the moment a site joins. Records stay where they were collected, so the established sites contribute no exemplars and need not remain reachable once the base model has been delivered. The arriving data carry no labels, so the emerging conditions have to be discovered before the output layer can grow. They are also scarce, since a recently joined site has little operating history and annotating it consumes the expert effort the federation was meant to conserve. What remains is one site, holding an unlabeled pool of tens of examples per condition, that has to extend a model other organizations depend on.

One representation carries all three stages. The base federated session leaves behind more than a classifier. Its early layers encode what the participating sites have in common, the structure of the underlying process rather than the classes that session contained, and they are identical at every site because the federation trained them together. We write ϕ_θ for the activation they produce and use it as the working space for every later stage, scoring novelty there, clustering the flagged records there, and describing old classes by their means in the same space. Since every stage consumes activations rather than measurements, what leaves a site is a summary and not a record.

Countering forgetting when the old data are elsewhere. Adding output units for the recovered conditions is mechanically simple. Keeping the model accurate on the classes it already had, while it trains on the new ones, is not. Something describing those classes has to enter the adaptation objective, and two sources remain once the established sites have gone offline. The federation can leave behind class prototypes, or the joining site can retain a small number of its own records, selected without labels by the same screen that identifies novelty. The two differ far less in what they achieve than in what they oblige an organization to disclose, so we treat them as a menu indexed by the governing privacy constraint rather than as an ablation ladder. Stated for a deployed system, the question is what a federation must carry forward in order to keep serving its members, and which of the available carriers a governance regime will permit.

Contributions.

1. An onboarding protocol for a site whose data are unlabeled and few, in which the incremental task is built autonomously at that site while the established sites stay offline, with screening, grouping and expansion all carried out inside one shared representation.
2. An evaluation on real ultrasonic-welding process data covering different patterns of class arrival, against distillation- and regularization-based continual learning.
3. A privacy-indexed account of old-class memory drawn from that evaluation. Federation-level prototypes and site-level records are dissimilar kinds of evidence, yet their accuracy differs by at most 0.022, so a deployment can choose between them by the disclosure it permits; both keep old-class and new-class accuracy within 0.026 of one another where every alternative we measure gives up one task to serve the other.

88 2 Related work

89 **Federated and federated class-incremental learning.** FedAvg established model averaging for
90 decentralized training [18], and later surveys catalogued the statistical, systems and privacy prob-
91 lems that heterogeneous participants create [11, 13]. Federated class-incremental learning admits
92 classes that arrive over time. Dong et al. [6] showed that local and global forgetting interact with
93 client heterogeneity, a further line treats the problem as one of selecting which participants’ knowl-
94 edge to accept rather than which exemplars to store [26], and a recent survey catalogues the task-
95 and class-incremental variants [25]. These protocols assume conditions the present setting does not
96 provide, namely labeled incremental tasks, clients that continue participating in federated rounds,
97 and a proxy server, per-client exemplar memory or server-side generator [6, 27]. Here a single or-
98 ganization arrives with an unlabeled pool and the established sites do not reconnect, so we compare
99 against the centralized continual-learning methods that the same literature adapts when a federated
100 protocol does not transfer [27].

101 **Open-set recognition and replay.** Open-set recognition requires a classifier to reject inputs out-
102 side its label set [2, 22], and reconstruction models suit the case in which examples of future classes
103 cannot exist at training time [5]. The federated version has been addressed through disagreement
104 between clients [29], which assumes the clients are online at the same time. We train one federated
105 expert per known class, and the same screen also selects the records that anchor the old classes.
106 Replay is among the most dependable defenses against forgetting [4, 17, 20], and its exemplars are
107 normally chosen using labels, whereas prototypes are summaries obtained by averaging.

108 **Discovering the classes before learning them.** A smaller body of work asks a federation to name
109 classes that no site has labeled. Federated generalized category discovery clusters unlabeled client
110 data into known and novel categories, but treats discovery as the end of the task rather than the
111 start of an expansion [19], and federated open-world semi-supervised learning spans both halves
112 with clients holding unlabeled data that contains novel classes [28]. Federated continual novel class
113 learning is the closest published setting to ours that we are aware of: its clients detect novel classes,
114 the server aligns their identity across sites, and training continues so that earlier classes survive
115 [23]. Two conditions separate our problem from it: discovery there proceeds while every site still
116 participates, whereas here the established sites are permanently offline and only a model and a small
117 set of class summaries outlive them; and its clients hold enough unlabeled data for cross-client
118 agreement to identify a novel class, while the site arriving here holds a few dozen records in total
119 and must decide alone. Few-shot class-incremental learning addresses the second condition in the
120 centralized case, typically by freezing the representation after a base session and placing new classes
121 with prototypes [24], but receives those classes already named and labeled.

122 **Cross-silo federations.** The setting studied here is cross-silo rather than cross-device, meaning
123 a small number of organizations, each holding data it is legally or contractually unable to release,
124 and each participating under a persistent identity [11, 13]. Hospitals, banks and manufacturers all
125 train shared models over records they cannot release to one another [1, 3, 7, 16, 21]. Two features
126 recur and shape the present problem: labeled data are scarce at any single silo, and the sensitivity
127 of the records is the reason the federation exists at all, so every quantity a site releases has to be
128 justified. We instantiate the protocol on ultrasonic welding, where unknown-fault detection and
129 few-shot adaptation have been studied for a single machine and a single dataset [8]. That setting
130 is centralized, so it faces neither the disclosure constraint nor the offline-participant constraint that
131 shape the problem here.

132 3 Problem formulation

133 3.1 Setting

134 Sites $1, \dots, M$ hold private datasets over a class set $\mathcal{Y}_{\text{old}}$ of size K , and a base federated session
135 learns f_{θ}^0 without moving raw records. After deployment, site $M+1$ joins with an unlabeled pool $\mathcal{U}$
136 containing a mixture of $\mathcal{Y}_{\text{old}}$ and a disjoint set $\mathcal{Y}_{\text{new}}$ whose size and identity are unknown on arrival.
137 The classifier expands to $K+R$ outputs using only what is available at the joining site together with
138 the summaries the federation retained.

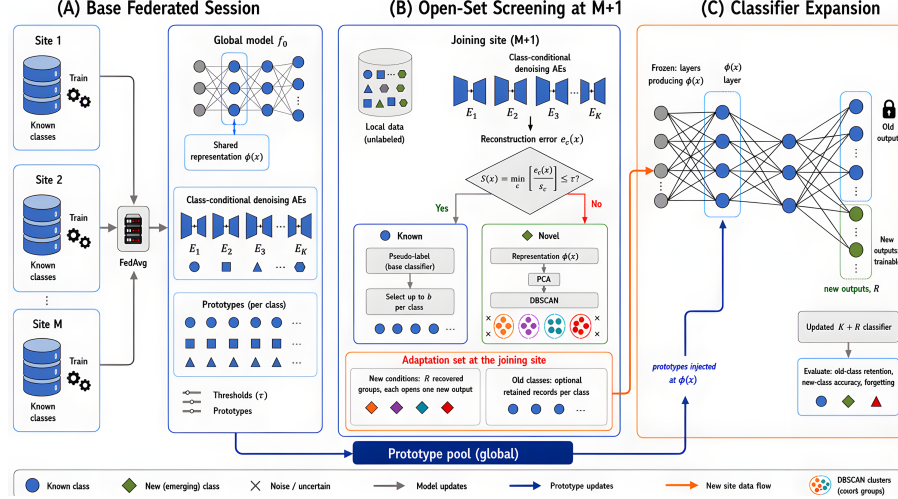


Figure 1: Onboarding a late-joining site. (A) The base federated session leaves behind a shared classifier, one reconstruction expert per known class, and representation-level prototypes. (B) The joining site screens its own unlabeled pool against those experts and clusters what it flags into candidate conditions. (C) The classifier expands by one output per recovered group, with the layers producing ϕ_θ held fixed. Prototype replay and local rehearsal are alternative sources of old-class evidence.

139 The federation retains the shared model, K reconstruction experts, the detection constants defined
 140 in Section 4, and one representation-level prototype per site and old class. The joining site keeps up
 141 to b of its own records per old class when the deployment permits it. Adaptation runs at site $M+1$
 142 alone, so the established sites need not reconnect and none of their records are transferred.

143 3.2 Evaluation protocol

144 We evaluate on one common held-out test set and report three quantities. Before the new site joins,
 145 A_{old}^0 is the old-class accuracy of the base model, averaged over the M established sites. After
 146 adaptation, A_{old} is the old-class accuracy of the expanded model averaged over those same M sites,
 147 and A_{new} is its accuracy on the conditions that arrived. There is one expanded model and one test
 148 set, and the per-site figures differ only in that each site reads those records through its own local
 149 input statistics. Forgetting is the drop on that same population, $F_{\text{old}} = A_{\text{old}}^0 - A_{\text{old}}$, so it measures
 150 what the established sites lose and the joining site never enters it. That site’s own old-class accuracy
 151 is reported separately, since no before-adaptation figure exists for it, and reporting the old classes
 152 before and after separates how much service the federation loses from how much of the new task it
 153 gains.

154 4 Method

155 The protocol performs three stages at the joining site. It screens the unlabeled pool for novelty,
 156 groups the flagged records into candidate conditions, and expands the classifier under an old-class
 157 memory constraint. Every stage operates on the same hidden representation, which is what allows
 158 the procedure to run without labels. Figure 1 identifies the quantities that cross a site boundary.

159 4.1 The shared representation

160 Each site trains a multilayer perceptron locally and the server averages the local parameters weighted
 161 by sample count. Let $\phi_\theta(x)$ be the activation of an early hidden layer of the resulting model. How
 162 many layers precede ϕ_θ is a hyperparameter of the protocol: deep enough that ϕ_θ no longer carries
 163 the input variables themselves, and shallow enough that what it encodes belongs to the process the
 164 sites share rather than to the classes the base session contained, which is what allows it to represent
 165 conditions absent at training time.

Each stage relies on a different property of this space. Screening needs one in which a record's dissimilarity to every known class can be measured, and the base session already shapes ϕ_θ so that the known classes separate in it. Grouping needs a space of manageable dimension in which records of one condition lie close together, and the same representation supplies one without a second embedding that the joining site has no labels to validate. Replay needs a form in which an established site can describe an old class without transmitting one of its records, and a mean activation serves, since every site reads this space through the same parameters while the mean itself is a summary rather than anything the site measured.

4.2 Screening: identifying unfamiliar records

For each $c \in \mathcal{Y}_{\text{old}}$ the established sites jointly train a denoising autoencoder g_c on representations labeled c only, using Gaussian input noise and the clean representation as the target. The server averages each expert independently. At inference,

$$e_c(x) = \|\phi_\theta(x) - g_c(\phi_\theta(x))\|_2^2, \quad S(x) = \min_{c \in \mathcal{Y}_{\text{old}}} \frac{e_c(x)}{s_c}. \quad (1)$$

The score has a simple interpretation. A familiar record is reconstructed well by one specialist, whereas an unfamiliar record is explained by none of them. Using one expert per class rather than a single global model is what makes this interpretation available, since an autoencoder fitted to all known classes averages over a multimodal distribution.

The divisor s_c puts the experts on a common scale. Classes are not equally easy to reconstruct, so an unnormalized minimum is governed by whichever expert fits most loosely. Site m reports the median error of expert c on its own class- c records, then a high quantile of its own known-class scores, and the server averages each across sites,

$$s_c = \frac{1}{M} \sum_{m=1}^M \text{med} \{e_c(x) : x \in \mathcal{D}_m, y = c\}, \quad \tau = \frac{1}{M} \sum_{m=1}^M Q_q(\{S(x) : x \in \mathcal{D}_m\}). \quad (2)$$

Each expert's error is thereby judged against what it normally achieves on the class it owns, and the joining site flags x when $S(x) > \tau$. Neither constant uses an example of a new class: each site releases K medians of its own known-class errors and one quantile of its own known-class scores, $K+1$ scalars beyond the model updates.

4.3 Grouping: estimating how many conditions have arrived

Screening identifies unfamiliar records but not how many conditions they represent. Their frozen representations are reduced by principal component analysis and clustered with DBSCAN [9], which needs no target cluster count and leaves low-density points unassigned. Each recovered group opens one new output, and no label is consulted.

4.4 Expansion and the evidence it requires

Once R groups exist, the output layer grows from K to $K+R$ units. The weight vector and bias that produce each of the K old-class logits are copied from the base model and their gradients are set to zero, so the linear map from the representation to the old outputs survives adaptation unchanged. Every configuration described below keeps this masking in place except FINETUNE, which is plain fine-tuning.

Preserving that map is not by itself enough to keep the old classes, because nothing in the adaptation objective refers to them. Containing forgetting needs evidence about the old classes inside that objective, and two sources remain available. They differ in what each obliges an organization to reveal.

Federation-level evidence: prototypes. Before the established sites disconnect, the base federation stores one prototype per site and old class,

$$p_{m,c} = \frac{1}{|\mathcal{D}_{m,c}|} \sum_{(x,y) \in \mathcal{D}_{m,c}} \phi_{\theta^0}(x), \quad (3)$$

Table 1: Adaptation configurations, what each holds fixed, and the disclosure each requires. b is the number of records the joining site retains per old class. Freezing ϕ_θ keeps prototypes and retained records in a coordinate system that does not shift while the classifier is expanded.

Name	Held fixed		Old-class evidence		Applicable when
	Old outputs	ϕ_θ	Prototypes	Records	
FINETUNE	no	no	no	no	no old-class evidence is available
FROZENHEAD	yes	no	no	no	only the model may be reused
GLOBAL	yes	yes	yes	no	class means may be released, nothing retained
LOCAL	yes	yes	no	b/class	nothing released, records retained locally
DUAL	yes	yes	yes	b/class	both disclosures are permitted

the mean activation of that site’s records for that class. The average is taken in the representation space deliberately. A mean of raw measurements would carry recognizable process values, whereas a mean activation lives in a space defined by parameters the federation already holds, so a site describes an old class without releasing anything it measured, and these MK vectors outlive the sites that produced them. During adaptation each prototype is injected at the frozen representation layer, forwarded through the remaining classifier, and its cross-entropy added to the loss on real data,

$$\mathcal{L}_{\text{inc}} = \mathcal{L}_{\text{CE}}(x, y) + \lambda_p \mathcal{L}_{\text{CE}}^{\text{proto}}(p_{m,c}, c). \quad (4)$$

Site-level evidence: local rehearsal. The joining site can instead assemble its own anchor set without labels, because the screen that identifies novelty also selects it. Records with $S(x) \leq \tau$ are pseudo-labeled by the frozen base classifier and the site retains up to b per predicted old class, drawn uniformly. These records are never transmitted and the established sites supply no class means.

4.5 A menu indexed by disclosure

Table 1 lists the configurations. They are not successive steps toward one best setting; each answers differently the question of what a deployment may release, and Section 6 reports the accuracy each obtains.

5 Experimental setup

Instantiation. The protocol is instantiated on ultrasonic metal welding, a joining process used in battery and electronics manufacturing. The data comprise 270 observations, one per welding cycle, each a vector of 252 engineered signal features recorded on production equipment, organized into nine balanced classes formed by three tool states (new, worn, damaged) crossed with three surface states. The new- and worn-tool combinations are the six old classes, and the three damaged-tool combinations arrive later. Tool damage is a realistic condition to arrive late, since it develops with use and need not have occurred anywhere in the federation while the base model was trained. It is also the demanding choice, since the arriving classes pair damaged tools with surface states the federation already knows and so differ from familiar conditions along one factor only.

The 30 records of each class are divided so that no observation is used twice. Six form a common test set on which no stage trains, six go to each of the three established sites for the old classes, and the remainder go to the joining site: six per old class and 24 per new class. The joining site therefore receives a pool of 36 familiar and 72 unfamiliar records, and every accuracy in the paper is measured on the 54-record test set. The data scale reflects the operating regime, since process data of this kind are proprietary, costly to label, and accumulate slowly at any one site.

Models and optimization. The base classifier is a multilayer perceptron with hidden widths 150, 90 and 40, trained for 20 federated rounds. In every experiment reported here ϕ_θ is the activation of the first of those layers and is therefore 150-dimensional. One denoising autoencoder per old class operates in that space, and the detection quantile is $q = 0.95$. Adaptation runs 20 site-local rounds with prototype loss weight one and $b = 3$ retained records per old class, so federation-level memory comprises $3 \times 6 = 18$ prototypes of dimension 150. Appendix E specifies the remaining hyperparameters.

Reference methods. Two established continual-learning methods are evaluated under the same architecture, split, optimization schedule and class-arrival patterns. LwF distills the frozen base model’s old-class outputs on the joining site’s adaptation records [14], and EWC penalizes parameter movement weighted by a diagonal Fisher information computed at the established sites and averaged by the server [12]. Both are centralized methods that the federated class-incremental literature adapts as references when its own protocols do not transfer [27]. Neither has a default weight that carries over here, so each is reported at the most favorable balanced setting it reaches in a sweep, which favors the reference methods; our own configurations use a fixed prototype weight of one and are not swept (Appendix B).

What we measure. We run two main experiments. The first isolates the expansion step. New classes are given with verified labels, so screening and grouping contribute no error and the only thing that varies is which old-class evidence a configuration replays. Every configuration is evaluated on all seven non-empty subsets of the three new classes, exercising arrivals of one, two and three classes, over 5 seeded splits, and Table 2(a) reports it. The second runs the procedure as deployed. The joining site’s pool is screened, grouped and used for expansion with no label consulted, so early errors propagate forward, and Table 2(b) reports it. Screening is also reported on its own, on the same pool.

A jointly trained nine-class model is not an available reference, since it would need simultaneous access to historical and joining-site records, so base-session accuracy serves as the reference instead. The two panels run separate base sessions, whose per-seed accuracies agree to within one test record. Reported spreads are sample standard deviations across seeds. Old-class accuracy is measured on the 36 old-class records of the test set, so one record is worth 0.028, and we do not read a difference below that as separating two configurations.

6 Results

6.1 Screening

On the pool the joining site receives, 36 familiar and 72 unfamiliar records, the detector reaches recall 0.975 ± 0.027 and acceptance 0.750 ± 0.068 at the federated threshold, with ROC-AUC 0.941 ± 0.018 ; on average it misses under two unfamiliar records and flags one familiar record in four in error. High recall is the right operating point for onboarding, since a condition that is never flagged is never learned, whereas a familiar record flagged in error is passed to grouping, where it is either left unassigned or absorbed into a recovered group. Appendix A compares this score against post-hoc detectors read off the same classifier.

6.2 The evidence that expansion requires

Table 2(a) attributes old-class accuracy to the evidence that produces it, averaging each seed over all seven patterns of class arrival and using verified new-class labels so that screening errors are excluded. Two configurations lose the old classes, fine-tuning falling to 0.560 ± 0.027 and FROZEN-HEAD reaching 0.655 ± 0.048 . Neither carries any description of the old classes into adaptation, and both forget accordingly.

Both forms of replay restore old-class accuracy to comparable levels. GLOBAL reaches 0.909 ± 0.048 , LOCAL reaches 0.924 ± 0.042 , and DUAL reaches 0.931 ± 0.035 with forgetting of 0.008 ± 0.020 . The three sit inside a band this experiment cannot resolve. The largest separation among them is 0.022, smaller than the 0.028 a single test record is worth, whereas each is separated from FROZENHEAD alone by 0.25 to 0.28, which is 9 test records or more. We therefore treat the three as equivalent in accuracy and separate them only by what they oblige a participant to disclose.

The lower block of panel (a) places these results beside established continual-learning practice. Both reference methods restrain movement away from the previous solution and control that restraint with a single scalar, so sweeping it moves accuracy along a frontier rather than raising it. At their balanced operating points LwF reaches 0.707 ± 0.039 old-class accuracy with 0.956 ± 0.023 on the new conditions, and EWC reaches 0.809 ± 0.054 with 0.935 ± 0.042 (Appendix B). The replay configurations lie on neither frontier, with DUAL sustaining 0.931 ± 0.035 and 0.905 ± 0.041 at once. Both reference methods obtain higher new-class accuracy and markedly lower old-class accuracy,

Table 2: Accuracy by configuration, over 5 splits. Old-class accuracy is given separately for the established sites and the joining site; the base session precedes the join and has no joining-site entry. Forgetting is the drop in established-site old-class accuracy, computed on those sites alone. (a) Controlled arrivals, averaged within each split over all seven arrival patterns; the lower block gives the baselines at their best regularization weight. (b) The full autonomous procedure.

(a) Controlled arrivals, verified new-class labels				
Configuration	Old (est. sites)	Old (joining)	New classes	Forgetting
Base session	0.939 $\pm$ 0.023			
FINETUNE	0.560 $\pm$ 0.027	0.567 $\pm$ 0.027	0.979 $\pm$ 0.010	0.379 $\pm$ 0.037
FROZENHEAD	0.655 $\pm$ 0.048	0.658 $\pm$ 0.045	0.979 $\pm$ 0.013	0.284 $\pm$ 0.060
GLOBAL	0.909 $\pm$ 0.048	0.913 $\pm$ 0.048	0.909 $\pm$ 0.060	0.030 $\pm$ 0.026
LOCAL	0.924 $\pm$ 0.042	0.927 $\pm$ 0.044	0.906 $\pm$ 0.039	0.015 $\pm$ 0.029
DUAL	0.931 $\pm$ 0.035	0.933 $\pm$ 0.035	0.905 $\pm$ 0.041	0.008 $\pm$ 0.020
LwF	0.707 $\pm$ 0.039	0.716 $\pm$ 0.037	0.956 $\pm$ 0.023	0.232 $\pm$ 0.042
EWC	0.809 $\pm$ 0.054	0.812 $\pm$ 0.054	0.935 $\pm$ 0.042	0.130 $\pm$ 0.041
(b) Full autonomous procedure				
Configuration	Old (est. sites)	Old (joining)	New classes	Forgetting
Base session	0.931 $\pm$ 0.025			
FINETUNE	0.149 $\pm$ 0.084	0.144 $\pm$ 0.087	0.944 $\pm$ 0.039	0.783 $\pm$ 0.066
FROZENHEAD	0.311 $\pm$ 0.170	0.311 $\pm$ 0.161	0.956 $\pm$ 0.025	0.620 $\pm$ 0.170
GLOBAL	0.872 $\pm$ 0.056	0.872 $\pm$ 0.064	0.878 $\pm$ 0.099	0.059 $\pm$ 0.059
LOCAL	0.868 $\pm$ 0.036	0.872 $\pm$ 0.037	0.900 $\pm$ 0.046	0.063 $\pm$ 0.037
DUAL	0.889 $\pm$ 0.055	0.889 $\pm$ 0.068	0.889 $\pm$ 0.039	0.043 $\pm$ 0.042

so the difference lies in retention rather than plasticity, and supplying evidence about the old classes is not equivalent to restraining movement away from them.

Replay also balances the two tasks more evenly. All three replay configurations keep old-class and new-class accuracy within 0.026 of one another, whereas every remaining entry in the table separates them by between 0.126 and 0.419. Appendix C varies the joining site’s data budget under LOCAL and finds the same balance sustained with 6 labeled records per arriving condition and 3 retained per old class.

6.3 The procedure end to end

Panel (b) of Table 2 chains every stage. The pool is screened, the flagged records are grouped, each group opens an output, and adaptation proceeds with no label consulted, so errors propagate into the new-class training set and, through pseudo-labeling, the old-class rehearsal buffer. A recovered group carries no name, so each is matched for reporting to the condition holding most of its members; in every split that match was one-to-one. Appendix D lists the grouping outcome of each split.

After autonomous expansion, DUAL retains 0.889 $\pm$ 0.055 on the old classes while reaching 0.889 $\pm$ 0.039 on conditions it discovered without supervision, against 0.149 $\pm$ 0.084 for fine-tuning. Retention is within a few points of the controlled setting despite the propagated errors, and the three replay configurations again fall inside one band, from 0.868 $\pm$ 0.036 to 0.889 $\pm$ 0.055. A rehearsal buffer assembled from predictions admits screening errors that prototypes computed before deployment do not, which is a reason to expect the two to diverge here, and these runs do not show it.

7 Selecting a configuration

The margins in Section 6 do not support reading Table 1 as an ablation ladder. What separates the replay configurations in practice is not accuracy but permission, and the two options ask permission of different parties. GLOBAL needs the established sites to release *MK* class means and needs nothing of the joining site, whereas LOCAL asks the established sites for no class means, needing only the model and the screening constants they publish once, and asks the joining site to retain a

few of its own records. Neither requirement implies the other, and a deployment ordinarily finds one foreclosed. Where an organization may publish representation-level statistics but the joining site is bound by a retention policy, GLOBAL applies. Where the established sites will release no class means, the common position between separate legal entities, LOCAL reaches comparable accuracy from records that never leave the joining site, and DUAL applies where both disclosures are permitted. A configuration can therefore be chosen from the constraint that actually binds, with its cost known in advance.

8 Limitations and future work

A fixed and common feature space. The protocol assumes that every site measures the same quantities in the same way and that this does not change while the federation runs. A prototype is interpretable at another site only because both compute ϕ_θ from the same inputs with the same parameters, and the threshold is comparable across sites for the same reason. Re-instrumenting a line, replacing a sensor or altering a feature-extraction pipeline breaks that correspondence, and a site arriving with a partly different feature set cannot be admitted as the procedure stands. Aligning heterogeneous feature spaces before the shared representation is formed is the extension we would attempt first.

Balanced classes and similar sites. Every class here contributes the same number of records and every site observes one process on one machine, so two pressures a real federation applies are absent. Rare conditions would arrive as a handful of records, moving the operating point of the screen, and sites differing by machine or operator would shift the representation the prototypes summarize.

Scope of the study. The federation is assumed to share one classification task with a fixed label space, and the protocol expands that space once per joining site. Sites carrying related but distinct tasks, and longer sequences in which the process drifts between arrivals, lie outside what we measure. Screening and grouping act on a representation and would carry over to other task families, whereas expansion is stated in terms of output units and would have to be restated for a regression or segmentation head, and the reported effect sizes are properties of one industrial instantiation. The reference methods are also compared as published, so neither receives the masking, freezing or retained records the replay configurations use. A fuller study would evaluate each stage against its own alternatives, screening, grouping and expansion in turn, and then assemble the strongest option at each stage into the autonomous procedure; that is the comparison we would run next.

9 Conclusion

We examined what a deployed federated model needs to absorb a site whose data are unlabeled and contain conditions no participant has observed. The protocol does this inside one shared representation, at the joining site, with no label consulted and no established site online. Containing forgetting requires evidence about the old classes inside the adaptation objective, and two dissimilar sources supply it at comparable accuracy, so a deployment can choose by disclosure rather than performance.

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A Screening

Screening is a supporting stage of the protocol, and the memory results do not depend on which detector supplies the candidate set. Table A1 compares the class-conditional reconstruction score against the two post-hoc scores that the joining site can compute from the shared model alone, the maximum softmax probability [10] and the energy score [15], neither of which needs a further statistic released by the established sites. All three are evaluated on the same 36 familiar and 72 unfamiliar records over 5 seeds.

Both classifier-confidence scores are weaker here, reaching 0.884 ± 0.033 and 0.875 ± 0.040 ROC-AUC against 0.941 ± 0.018 for the reconstruction bank, which is consistent with a six-class softmax trained on 36 observations per site being poorly calibrated. The margins, 0.057 and 0.066, are larger than any separation among the memory configurations of Table 2, and the bank ranks ahead of the energy score on 5 of 5 splits and ahead of maximum softmax probability on 4 of 5. What the bank adds beyond the margin is a per-class account of why a record was flagged, since the errors of the individual experts are available alongside the decision.

Table A1: Post-hoc screening scores on the joining site’s pool, over 5 seeds. Recall is measured with each score thresholded at the 0.9 quantile of its own familiar-record scores, so all three operate at the same acceptance rate, 0.89. Mean $\pm$ sample standard deviation.

Score	ROC-AUC	Average precision	Recall
Maximum softmax probability	0.884 ± 0.033	0.923 ± 0.032	0.614 ± 0.085
Energy score	0.875 ± 0.040	0.932 ± 0.020	0.636 ± 0.095
Class-conditional reconstruction (ours)	0.941 ± 0.018	0.950 ± 0.034	0.872 ± 0.057

B Reference-method sweeps

Figure A1 reports the full regularization sweep for both reference methods, 12 weights for LWF spanning 0.05 to 200 and 6 for EWC spanning 10^8 to 10^{14} , each averaged over 5 seeds. Both were extended until the harmonic mean of the two accuracies peaked and began to decline, so neither method is reported at an under-tuned setting; the rows in Table 2 correspond to weights 0.25 and 10^{11} . That balanced point favors the reference methods, which are the only entries in the table selected this way; our own configurations use a fixed prototype weight and are not swept.

The sweep is presented rather than a single row because any choice of operating point is arguable, and the figure makes the comparison independent of that choice. The replay configurations sit above both sweeps at every weight. At comparable new-class accuracy they retain substantially more, and no setting of either weight reaches the region they occupy.

C How much data the joining site needs

Both kinds of evidence that adaptation consumes come from the joining site, namely the labeled records that define the arriving conditions and the records it retains for the old ones. Neither is plentiful at a site that has recently started operating, so the question a deployment asks is not how the protocol behaves at the budget used in Section 5 but how far that budget can be cut. We answer it in the LOCAL configuration, where no prototype is available and every training record therefore comes from the joining site, over a grid of 25 budgets: n from 1 to 6 labeled records per arriving condition, and b from 0 to 6 retained records per old class. All three conditions arrive at once, every other setting is that of Section 5, and each cell is averaged over 5 splits.

Figure A2 reports the grid, and the two budgets separate along the two axes.

Arriving-condition records buy plasticity. New-class accuracy, panel (b), is governed by n and is nearly flat in b . One labeled record per condition leaves it at 0.556 ± 0.104 at the default $b = 3$, since a single point fixes an output unit without describing the region around it. Three records reach 0.800 ± 0.115 and six reach 0.900 ± 0.046 , so most of the plasticity the site needs is bought within the first handful of examples.

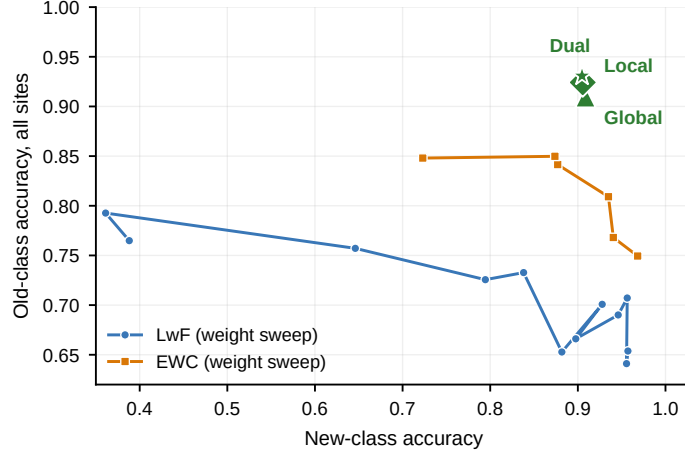


Figure A1: Regularization sweeps for the reference methods against the replay-based configurations, 5 seeds. Each curve traces one method as its regularization weight increases. Old-class accuracy is bought with new-class accuracy along each sweep; the replay configurations lie on neither.

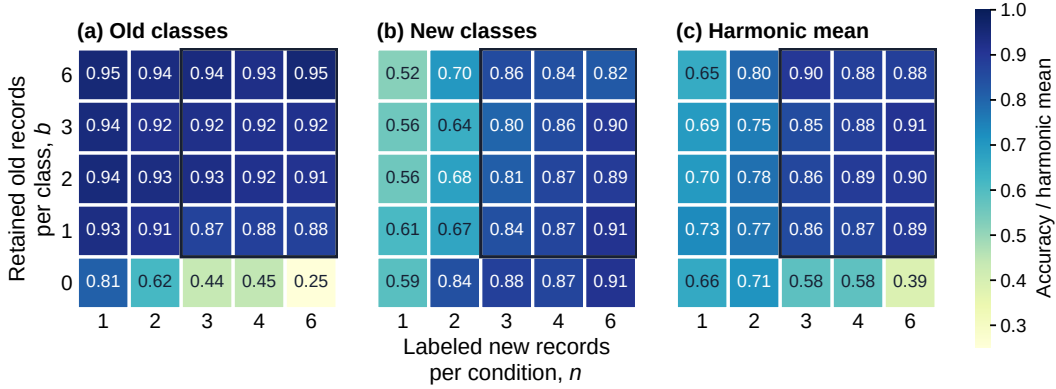


Figure A2: Joining-site data budget under LOCAL, over 5 splits with all three conditions arriving at once. n is the number of labeled records per arriving condition and b the number retained per old class. (a) old-class accuracy over all participating sites and (b) new-class accuracy at the joining site are the two quantities Table 2 reports; (c) is their harmonic mean, which is high only where both are. The outlined block marks the region $n \geq 3$, $b \geq 1$ discussed in the text.

487 **Retained records buy retention, and the first one does most of it.** Old-class accuracy, panel (a),
488 is governed by b . The $b = 0$ row is adaptation with no old-class evidence at all, and it is the only
489 region of the grid where the model loses the old classes, averaging 0.513 ± 0.209 across that row. It
490 also degrades as n grows, falling to 0.253 ± 0.083 at $n = 6$, which is the mechanism of Section 6
491 seen directly: the more the site trains on the arriving conditions, the further the shared layers move
492 and the less the preserved output map means. A single retained record per class lifts that column to
493 0.876 ± 0.053 , a second reaches 0.910 ± 0.060 , and the rows above change little.

494 **The cheapest adequate budget.** Panel (c) is high only where both accuracies are, and it shows a
495 broad plateau rather than a single optimum. Every cell with $n \geq 3$ and $b \geq 1$ lies within 0.054 of the
496 best cell in the grid, $n = 6$ with $b = 3$, at 0.917 ± 0.065 on the old classes and 0.900 ± 0.046 on the
497 new ones. Outside that region one of the two gives way, the old classes at $b = 0$ and the new ones
498 below $n = 3$. The budget the main experiments use is far larger, $n = 24$ and $b = 3$, and reaches
499 0.889 ± 0.059 and 0.978 ± 0.030 under the same configuration and splits, so the largest cell in the
500 grid is already above the full-pool budget on the old classes and below it on the new ones. Reducing
501 the arriving-condition budget costs accuracy on the conditions the site is itself trying to learn and

returns a little on the classes the federation already served, which is the more forgiving of the two directions for a deployment to be wrong in. This is also the practical reading of the disclosure argument in Section 7. A retention policy that permits one or two records per old class, together with an operating history that has produced three examples of each new condition, is enough to admit the site.

D Per-split grouping outcomes

Table A2 reports the grouping stage for each of the 5 complete runs in Table 2(b). “Groups recovered” is the number of conditions the clustering returned, and the partition agreement with the reference conditions varies across splits. “Unassigned” counts flagged records that DBSCAN left in no group; these are excluded from the new-class training set, which is why a run can recover the correct number of groups and still train on fewer records than it flagged.

Table A2: Grouping outcomes for the complete runs. ARI is the adjusted Rand index between the recovered partition and the reference conditions, computed on the assigned records.

Seed	Groups recovered	ARI	Unassigned
7	3	0.764	12
11	3	0.507	9
42	3	0.734	11
123	3	0.679	13
314	3	0.667	9

E Reproducibility and compute

Experiments ran with Python 3.11, TensorFlow 2.15.1 (Keras 2.15), NumPy 1.26.4, pandas 1.5.3, scikit-learn 1.3.2, and Matplotlib 3.7.5, on two CPU cores with no GPU. One complete chained run covering all configurations takes roughly ten minutes, one controlled sweep over the seven subsets about five, and one seed of the reference-method sweep about five, and one seed of the budget grid in Appendix C about six. We re-ran the base federated session after every environment change and obtained identical end-of-session accuracy, with later stages agreeing to within one test observation.

Randomness is controlled for NumPy and TensorFlow operations. Reported variation is the sample standard deviation across independently seeded runs, used descriptively; we draw no inferential conclusion from it.

The base classifier uses ReLU activations and a softmax output and is trained for 20 federated rounds with one local epoch per round, batch size four, and Adam at 10^{-3} . Each denoising autoencoder has hidden width 64, latent width 32, Gaussian input noise of standard deviation 0.02, and ℓ_2 coefficient 10^{-4} , trained for 30 federated rounds with two local epochs. Grouping uses four principal components, and DBSCAN six minimum neighbors with a radius at the 60th percentile of empirical nearest-neighbor distances. Adaptation runs at batch size four and learning rate 10^{-3} .

Symbol	Meaning
M, K	number of established sites; number of old classes
R	number of new conditions recovered and added as outputs
$\mathcal{U}$	unlabeled pool held by the joining site
$\phi_\theta(x)$	early-layer activation, shared by all stages
$g_c, e_c(x)$	reconstruction expert for class c ; its squared error
s_c	scale of expert c , its median error on its own class
$S(x)$	screening score, $\min_c e_c(x)/s_c$
τ, q	federated threshold; detection quantile
$p_{m,c}$	prototype of class c at established site m
b	local rehearsal records kept per old class
λ_p	weight on the prototype replay loss
$A_{\text{old}}^0, A_{\text{old}}$	old-class accuracy before and after integration
A_{new}	new-class accuracy after integration
F_{old}	forgetting, $A_{\text{old}}^0 - A_{\text{old}}$

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